

TEST PLAN TEMPLATE (ISTQB v4.0)

Detailed Guidance for Each Section

DOCUMENT INFORMATION

Field	Description
Project Name	Enter the official project/system name
Document Title	TEST PLAN
Version	Example: v1.0
Prepared By	QA Lead / Tester Name
Reviewed By	Project Manager / QA Manager
Approved By	Stakeholder / Customer
Date	Document creation date
Status	Draft / Review / Approved

REVISION HISTORY

Version	Date	Author	Description
1.0			Initial version
1.1			Updated test scope

1. INTRODUCTION

Purpose

Describe why this Test Plan document exists.

What to write:

- Objective of testing

- Business value of testing
- Overview of the system
- Why testing is important for this project

Example:

This Test Plan defines the testing approach, scope, schedule, resources, tools, and activities for the English Center Management System – Multi-Branch Version. The objective is to ensure the software meets functional and non-functional requirements before production release.

2. PROJECT OVERVIEW

Purpose

Provide a brief summary of the project.

What to write:

- System overview
- Main business processes
- Key modules
- Technologies used
- Number of users/branches

Example:

The system supports management of multiple English center branches including student management, class scheduling, teacher assignment, attendance tracking, tuition management, and reporting.

Recommended content:

- Frontend: Vue 3
 - Backend: .NET 9 API
 - Database: PostgreSQL
 - Authentication: JWT
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3. TEST OBJECTIVES

Purpose

Define the goals of testing.

What to write:

- Verify functional requirements
- Validate business workflows
- Detect defects early
- Ensure system stability
- Verify security and performance

Example:

- Ensure all critical business functions operate correctly.
 - Verify role-based access control.
 - Ensure APIs respond within acceptable performance limits.
 - Validate integration between frontend and backend.
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4. TEST SCOPE

Purpose

Define what will and will not be tested.

4.1 In Scope

What to write:

List all modules/features included in testing.

Example:

Module	Description
Authentication	Login, logout, forgot password
Student Management	CRUD student information
Class Management	Schedule and classroom management
Teacher Management	Assign teachers to classes
Tuition Management	Payment and invoice processing
Reports	Revenue and attendance reports

4.2 Out of Scope

What to write:

Features excluded from current testing.

Example:

- Third-party CRM integration
 - AI facial recognition attendance module
 - Mobile application testing
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5. TEST ITEMS

Purpose

Identify software items/components to be tested.

What to write:

- Modules
- APIs
- UI screens
- Databases
- Reports
- External integrations

Example:

Item	Description
Login API	User authentication API
Student Screen	Student management page
PostgreSQL DB	Database validation

6. TEST TYPES

Purpose

Specify types of testing to perform.

Recommended test types:

Test Type	Purpose
Functional Testing	Validate business functions
Integration Testing	Verify module communication
System Testing	Validate end-to-end workflows
Regression Testing	Ensure changes do not break existing features
Performance Testing	Evaluate speed and scalability
Security Testing	Validate authentication and authorization
Usability Testing	Verify user friendliness
Compatibility Testing	Browser/device compatibility

Example:

Regression testing will be executed after every sprint release.

7. TEST APPROACH / TEST STRATEGY

Purpose

Explain how testing will be conducted.

What to write:

- Manual or automation testing
- Black-box / White-box techniques
- Risk-based testing
- Agile testing approach
- Test levels

Example:

Testing Level	Approach
Unit Testing	Performed by developers using NUnit
Integration Testing	API testing with Postman
System Testing	Manual testing using test cases
Automation Testing	Selenium for regression testing

Techniques:

- Equivalence Partitioning
 - Boundary Value Analysis
 - Decision Table Testing
 - Use Case Testing
 - State Transition Testing
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8. ENTRY CRITERIA

Purpose

Define conditions required before testing begins.

Example:

- Requirements approved
 - Test environment ready
 - Build successfully deployed
 - Test data prepared
 - Test cases reviewed
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9. EXIT CRITERIA

Purpose

Define conditions required to stop testing.

Example:

- 100% critical test cases executed
 - No Severity 1 defects remain open
 - Test summary report completed
 - Customer UAT approved
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10. TEST ENVIRONMENT

Purpose

Describe the testing infrastructure.

What to write:

- Hardware
- Software
- OS
- Browsers
- Servers
- Databases
- Devices

Example:

Component	Configuration
Frontend	Vue 3 + PrimeVue
Backend	.NET 9 Web API
Database	PostgreSQL 16
Browser	Chrome, Edge, Firefox
OS	Windows Server 2022

11. TEST TOOLS

Purpose

List tools used during testing.

Example:

Tool	Purpose
Jira	Defect tracking
Selenium	Automation testing
Postman	API testing
JMeter	Performance testing
GitHub Actions	CI/CD execution
Katalon	UI automation testing

12. TEST DATA MANAGEMENT

Purpose

Describe how test data will be created and maintained.

Example:

- Use anonymized production data
- Generate mock student records
- Prepare separate datasets for positive and negative testing

Sample datasets:

Dataset	Purpose
Valid Accounts	Positive login testing
Invalid Accounts	Negative testing
Large Dataset	Performance testing

13. ROLES AND RESPONSIBILITIES

Purpose

Define responsibilities for each team member.

Example:

Role	Responsibility
QA Lead	Manage test planning and reporting
Tester	Execute test cases and log defects
Developer	Fix defects
Project Manager	Approve deliverables
Customer	Perform UAT

14. TEST SCHEDULE

Purpose

Define testing timeline.

Example:

Phase	Start Date	End Date	Owner
Test Planning			
Test Case Design			
Environment Setup			
Test Execution			
Regression Testing			
UAT			
Test Closure			

15. DEFECT MANAGEMENT PROCESS

Purpose

Describe how defects are reported and tracked.

Example workflow:

New → Assigned → In Progress → Fixed → Retest → Closed

Severity levels:

Severity	Description
Critical	System crash/data loss
High	Major business function failure
Medium	Partial function issue
Low	Cosmetic/UI issue

16. RISKS AND MITIGATION

Purpose

Identify project risks and mitigation plans.

Example:

Risk	Impact	Mitigation
Delayed build delivery	Testing delay	Daily build monitoring
Unstable environment	Failed executions	Dedicated QA environment
Lack of test data	Low coverage	Create mock datasets

17. TEST DELIVERABLES

Purpose

List documents generated during testing.

Example:

- Test Plan
 - Test Cases
 - Test Scripts
 - Defect Reports
 - Test Execution Logs
 - Test Summary Report
 - RTM (Requirement Traceability Matrix)
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18. COMMUNICATION PLAN

Purpose

Define communication channels and reporting frequency.

Example:

Activity	Frequency	Participants
Daily QA Meeting	Daily	QA Team
Defect Review Meeting	Twice per week	QA + Dev
Test Status Report	Weekly	Stakeholders

19. APPROVALS

Name	Role	Signature	Date
	QA Lead		
	Project Manager		
	Customer Representative		

APPENDIX A – TEST CASE TEMPLATE

Field	Description
Test Case ID	Unique identifier
Module	Functional area
Preconditions	Required conditions
Test Steps	Execution steps
Expected Result	Expected system behavior
Actual Result	Real result
Status	Pass/Fail

APPENDIX B – DEFECT REPORT TEMPLATE

Field	Description
Defect ID	Unique defect identifier

Field	Description
Summary	Short defect description
Steps to Reproduce	Reproduction steps
Expected Result	Expected behavior
Actual Result	Actual behavior
Severity	Critical/High/Medium/Low
Priority	P1/P2/P3
Status	Open/Fixed/Closed

APPENDIX C – DECISION TABLE TESTING SAMPLE

Conditions	TC1	TC2	TC3	TC4
Valid Username	Y	Y	N	N
Valid Password	Y	N	Y	N
Expected Result	Login Success	Invalid Password	Invalid Username	Login Failed

APPENDIX D – USE CASE TESTING SAMPLE

Use Case: Add Student

Step	User Action	Expected Result
1	Open Student Page	Student list displayed
2	Click Add New	Add form displayed
3	Enter valid information	Data accepted
4	Click Save	Student saved successfully

END OF DOCUMENT